

VADI VH-3000 MANUAL



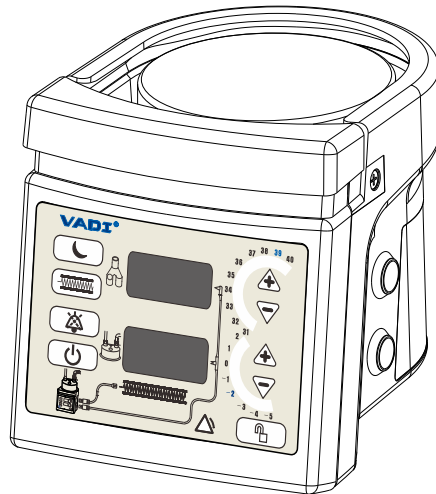
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VADI

VH-3000

Respiratory Humidifier



CE 2460

Manual

Read instruction carefully before use

2021.12.23

Made In Taiwan
VA-1-VH-3000-EN.MDR

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1. General Information



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Borkstrasse 10, 48163 Münster, Germany



MedNet SWISS GmbH

Bäderstrasse 18, 5400 Baden, Switzerland

2. Specification and Definitions

Dimension: 17.5cmX13.5cmX14cm

Weight: 1500g

I/P: 100-120 Vac, 50/60Hz Sine Wave, 3.4A or

200-240 Vac, 50/60Hz Sine Wave, 1.7A

(Please refer to the mark on device.)

Fuse Rating: T4AL/250Vac

Heated Wire Power Supply: DC24V / 120 watts Max

Overheating Protection:

While heated temperature is over $120^{\circ}\text{C} \pm 7^{\circ}\text{C}$, the device will cut out the power automatically.

The VH-3000 life cycle is 3 years.

Operating Environment Temperature:

18~26°C, Humidity RH:15%~95%, Pressure: 700 to 1060hPa

Storage Temperature: 25~40°C, Humidity RH:15%~95%

without condensation

Transport Temperature: -10~60°C, Humidity RH:15%~95%

without condensation, Pressure: 595 to 1013 hPa

Warm-up Time: ≤ 30 minutes

Suggested Flow Range: 2~60 LPM

Humidity Output :>33mg/l within recommended flow range

Standard Compliance: IEC/EN/ES 60601-1, IEC/EN 60601-1-2,
IEC 60601-1-8, ISO 80601-2-74

Alarm Sound Pressure: > 45dB

Classified as: Class I, Type BF, IPX 1

Symbols Glossary

Symbol Title and Meaning

	Manufacturer
	Date of manufacture
	Consult instructions for use
	Medical device
	Temperature limit
	Storage Humidity limitation
	Batch code
	Catalogue number
	Unique device identifier
	EU Authorized Representative
	Switzerland Authorized Representative
	Use-by date
	Serial number
	TYPE BF Applied Part (inhalation tube)
	Caution Electrical Shock Hazard. Refer to qualified service Personnel.
	Caution Hot surface may exceed 75°C
	Drip-proof
	Alarm
	Silence

	CE Mark
	Power On/Off (Standby)
	Protective earth
	Caution
	Fragile, handle with care
	Keep dry
	Use no hooks
	Handle with care
	This side up

3. Intended Use

A respiratory humidifier is a device that is intended to heat and warm breathing gases before they enter a patient's airway.

Notes

- 1.The use of breathing circuits, chambers or other accessories which are not from by VADI Medical Technology may affect the performance or compromise safety.
- 2.Ensure maintenance of grounding integrity by connection to a "hospital grade" receptacle.
- 3.Always disconnect power supply before servicing.
- 4.When installing a humidifier close to a patient, the humidifier is always securely mounted and positioned lower than the patient.
- 5.Please keep the humidifier away from the high frequency surgical apparatus, shortwave or microwave equipment. It may affect its performance.
- 6.Ensure the temperature probe sensors correctly and securely fitted. Failure to do so may result in gas temperature in excess of 41°C being delivered to the patient.
- 7.Do not touch the tip of the chamber temperature probe during use.
- 8.Visually inspect accessories free from damage before use.
- 9.Normal operation cannot be guaranteed if power from a source other than a pure sine wave, such as a square wave inverter.
- 10.The VH-3000 can be used in all age which is upper airway malfunction, include in neonatal, child, and adult.
- 11.The VH-3000 can be applied on any ventilator which include in servo control, invasive, maximal flow rate 60 LPM.
- 12.The VH-3000 accessories can comply adult (ID/OD 22 mm), child (ID/OD 15 mm), neonate (ID/OD 12 mm) ventilator connector.
- 13.Warning against servicing and maintenance while the ME EQUIPMENT is in use.

The reprocessing of the accessories:

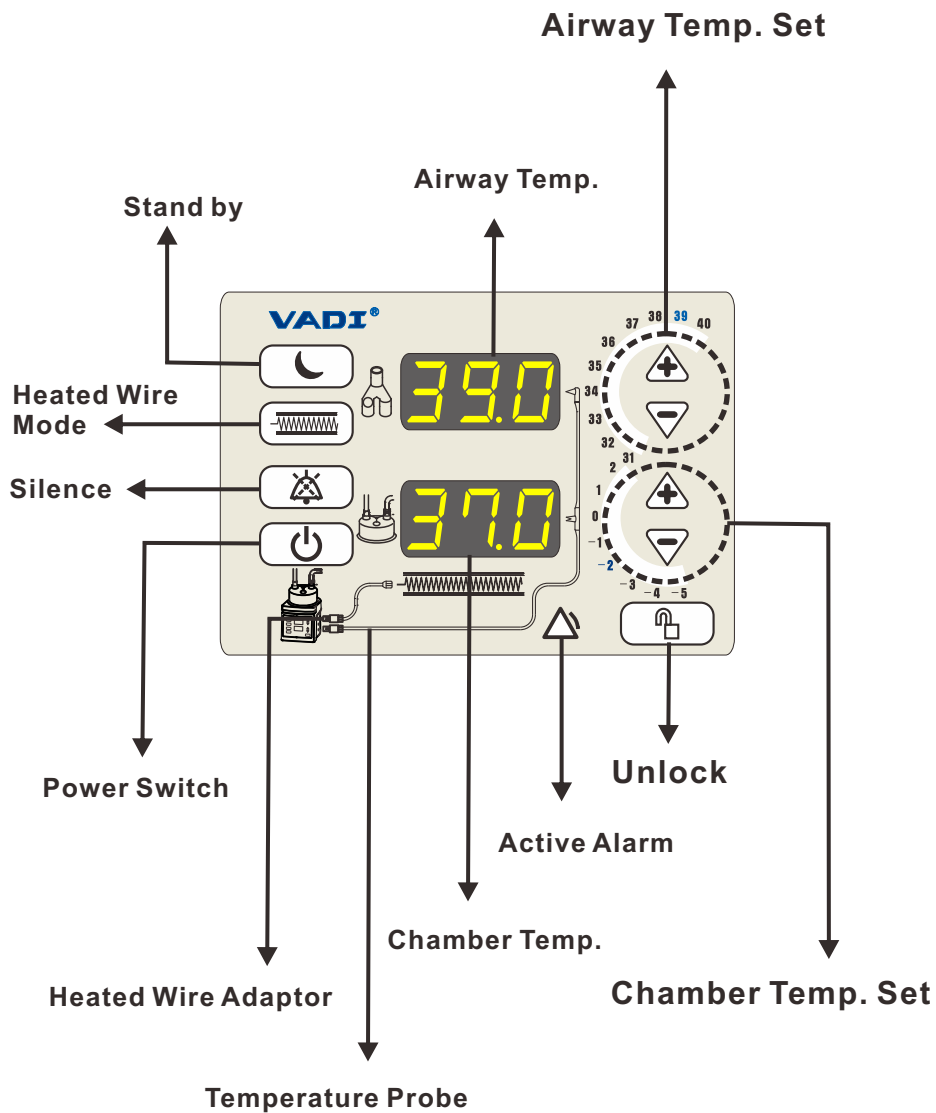
- Breathing circuit:

1. Dismantle all components except the stainless steel spring attached to the body of water trap.
2. Clean the parts with low flow running water, using a soft brush to remove all foreign material. (for example, soil or organic matter)
3. Sterilization by autoclave: 132°C/ 4minutes.

- Chamber:

1. Clean the parts with low flow running water, using a soft brush to remove all foreign material. (for example, soil or organic matter)
2. Sterilization by autoclave: 132°C/ 4minutes.

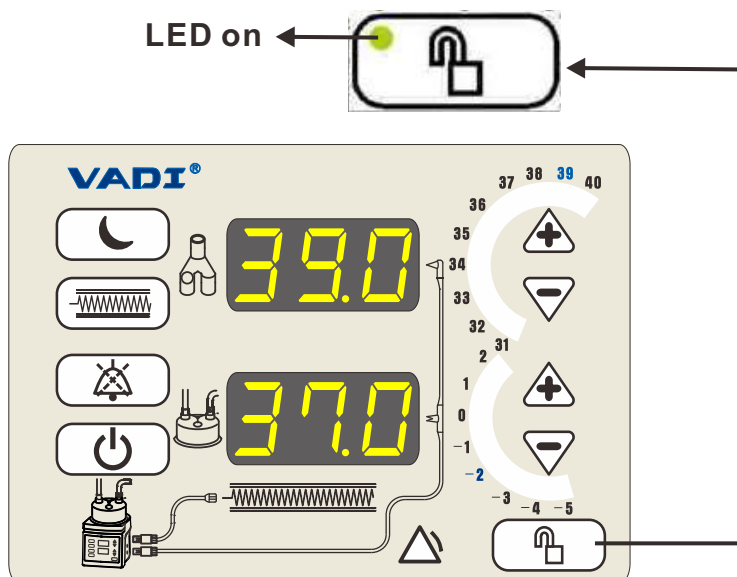
4. Control Panel



5. Function Introduction

5.1 Unlock

Press “unlock” until the LED is on, that users can operate the humidifier. The unlock button will automatically resume if no action is taken within 10 sec.

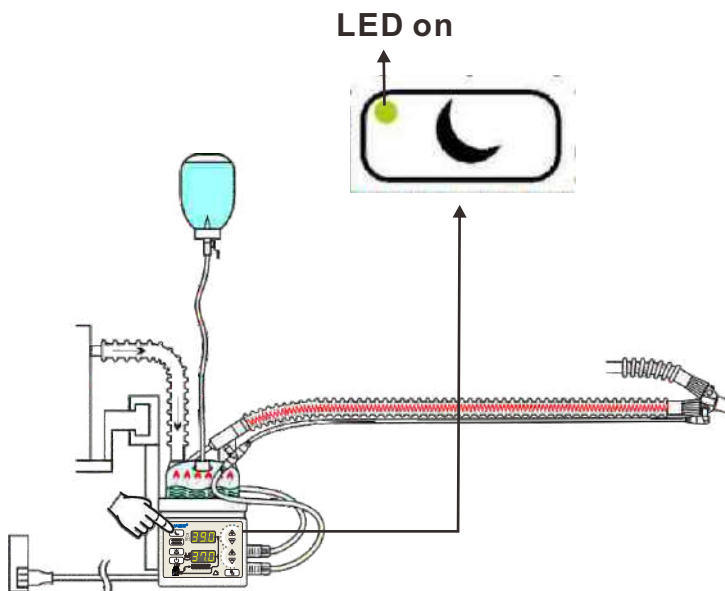


※The following buttons are locked if the user doesn't activate the Unlock button.

1. Standby
2. Heated Wire/Non-Heated Wire Mode
3. Power Switch (Except the first power on)
4. Airway Temp. Set
5. Chamber Temp. Set

5.2 Standby

If the button is pressed, the LED is on. Then 40°C on heater plate and 20% power supply to the heated wire are kept for 15 minutes. Push the button again to disable the function.

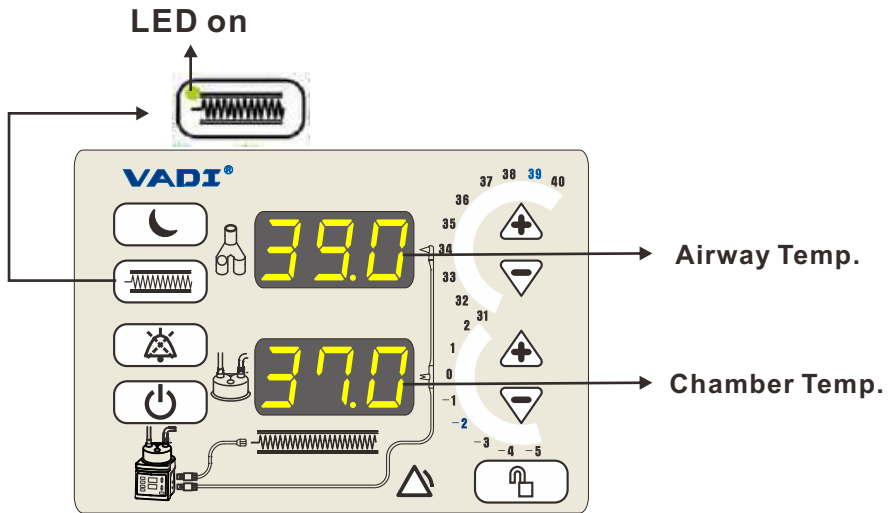


Notes

1. In standby mode, it will not show "No FLo".
2. The humidifier will resume the period of standby mode if the power is pressed again within 15 minutes after switched off. But power failure will not resume the function.

5.3 Heated Wire Mode

While the LED is on, the function is activated. If the function is off, it turns to Non-Heated Wire Mode.



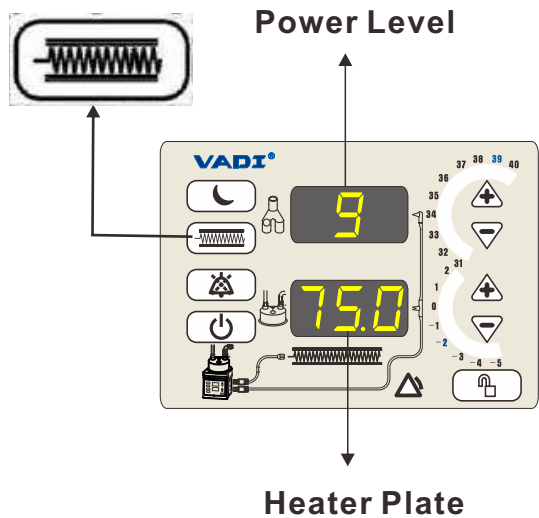
The user can set the airway and chamber temp. via adjustment of “+” or “-” button.

EX: If airway temp. set is 39°C and chamber temp. set is -2°C, the humidifier maintains 39°C at airway and 37°C at chamber.

Note : The default is 39°C at airway temp. set and -2°C at chamber temp. set.

5.4 Non-Heated Wire Mode

While the LED off, the function is activated.



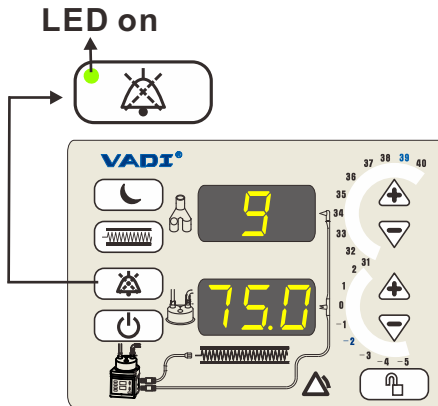
Power Level	1	2	3	4	5	6	7	8	9
Heater plate Temperature	45°C	48°C	52°C	56°C	60°C	64°C	68°C	71°C	75°C

The user use “+” or “-” button on airway temp. set to increase or decrease the Power level.

5.5 Silence

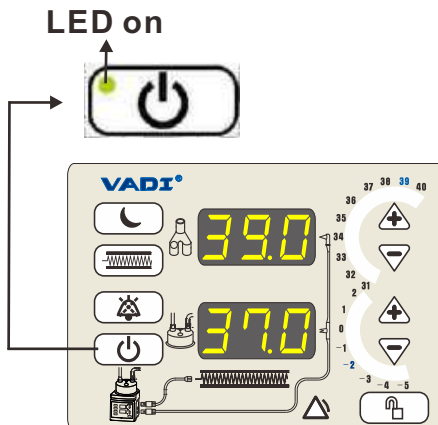
The button is used to turn off audio alarm for 120 sec.

When the airway temp. exceeds 42°C, Silence is limited to 30 sec.



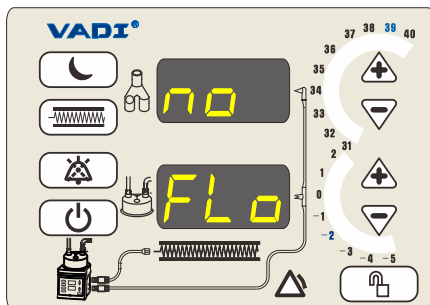
5.6 Power Switch

It is on/off switch. The LED is on while the humidifier is activated.



5.7 NO FLO

When airway temperature fails to reach 28°C and no increase in 30 Seconds, the LED displays "no Flo".



5.8 Hi / Lo Alarm

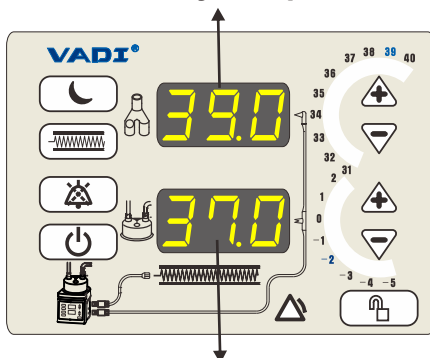
If it is lower than 10°C , it displays "Lo".

If it is higher than 70°C, it displays "Hi".

Airway temp. : It is the temperature of Airway.



Airway temp.



Chamber Temp.

Chamber Temp. : It is the temperature of Chamber.

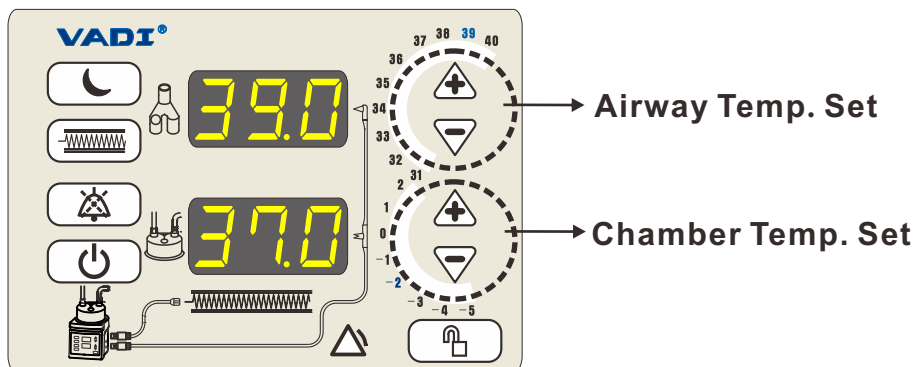


5.9 Temperature Set

(A) Airway Temp. Set

The LED on Airway Temp. Set should be kept the same as the previous one while being switched on.

※The user can adjust it by “+” and “-” from 31°C to 40°C.

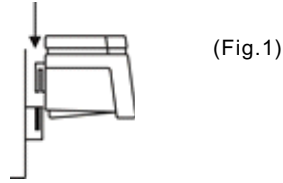


(B) Chamber Temp. Set

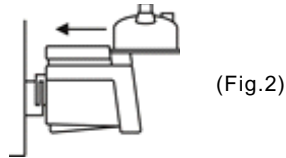
The user can adjust it by “+” and “-” from -5°C to +2°C.

6. Installation

6.1 Slide the device on the bracket .
(Fig.1)



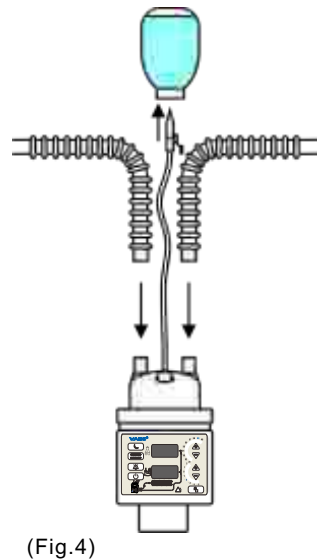
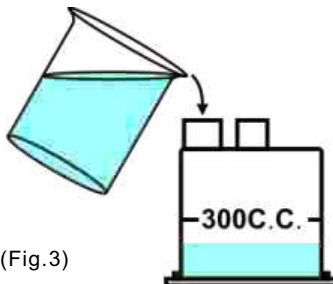
6.2 Slide chamber onto the heater base.
(Fig.2)



Note

-Choose the appropriate chamber.

6.3 Add distilled water into chamber.(Fig.3) or stab distilled water with the needle of IV-SET and connect the tube to humidification chamber.(Fig.4)



6.4 Connect a breathing circuit to the chamber.

6.5 Connect temperature probe to humidifier, chamber and inspiratory tube.

6.6 Connect the heated wire cable to the circuit.

Notes

1. DO NOT fill the chamber above the maximum fill level line. Liquid could enter the breathing circuit if the chamber is overfilled.
2. When mounting a humidifier adjacent to patient, ensure that the humidifier is always positioned lower than the patient.
3. Ensure that both temperature probe sensor are correctly and securely fitted. Failure to do so may result in temperatures in excess of 41°C being delivered to the patient.

7. Use Instruction

7.1 Be sure of correct installation of chamber and breathing circuit.

Fig.1 Heated Wire Breathing circuit System

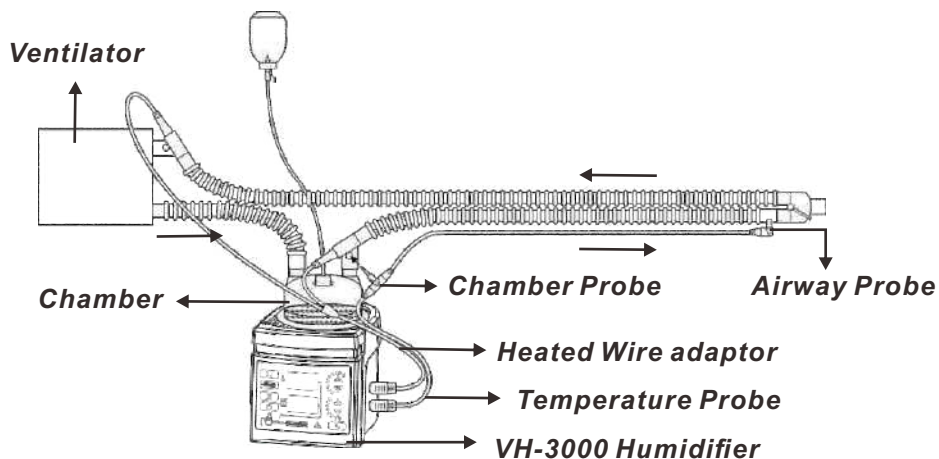
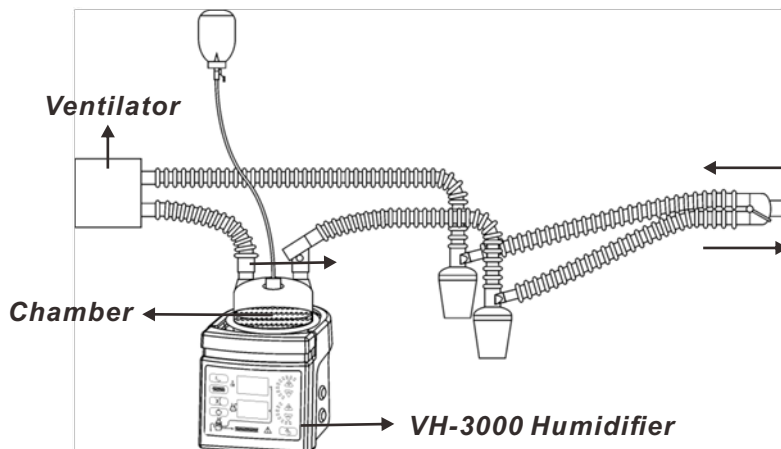


Fig.2 Non-heated wire breathing circuit system



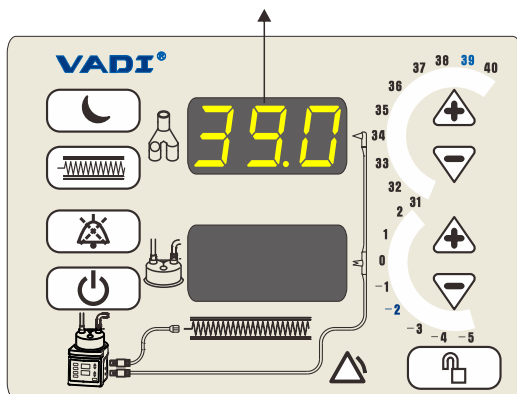
- 7.2 Plug the humidifier power cable into an AC supply of the voltage and maximum power rating specified on the side label of the unit.
- 7.3 Ensure that the humidification chamber and breathing circuit are installed and connected correctly, refer to Section 6.
- 7.4 Switch on the ventilator or gas supply. Carry out flow and pressure testing applicable to the ventilator in use.
- 7.5 Set the temperature level refer to 5.1, 5.3, 5.9
- 7.6 If a heated wire circuit is being used, check that the green indicator LED next to the button labeled “heated wire” is illuminated. If not, press the button firmly. Ensure that the green LED remains illuminated when the button is released.
- 7.7 If an alarm is activated, refer to Section 8

Notes

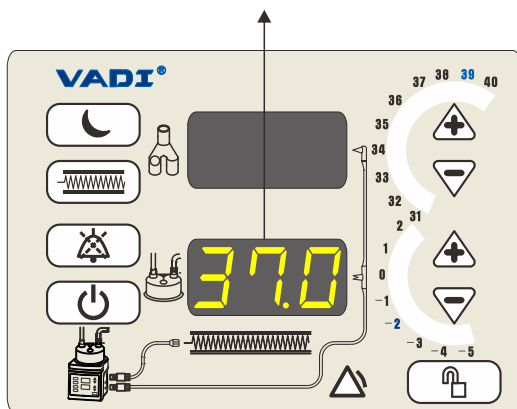
1. Breathing circuits should be changed according to local clinical guideline to avoid infection.
2. Avoid the leaning of the device more than 10°. Keep humidifier horizontally.
3. Do not touch the heater plate to avoid thermal burns.

8. Alarms

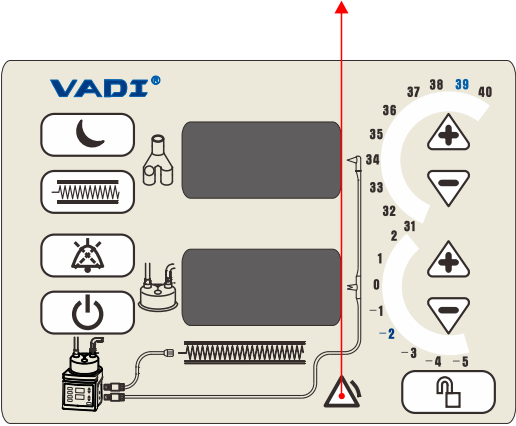
8.1 The alarm tolerance of Airway Temp is 2°C above or 4°C lower than the temperature control setting, The Airway Temp LED will flash and audible alarm.



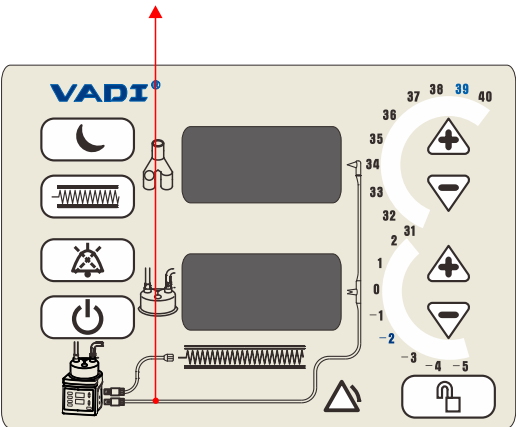
8.2 The alarm tolerance of Chamber Temp. is above or lower than 4°C the temperature control setting, The Chamber Temp. LED will flash and audible alarm.



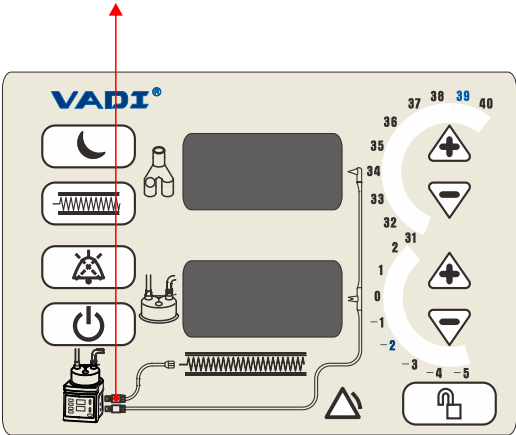
8.3 If the exclamation flash and audible alarm, please contact your local VADI distributor.



8.4 The Temperature probe disconnect or malfunction, the LED will flash and audible alarm.



8.5 The heated wire adaptor disconnect or malfunction, the LED will flash and audible alarm.



9. Troubleshooting

Alarm Priority: MID

ALARM CONDITIONS	REQUIRED ACTION
The alarm tolerance of Airway Temp is 2°C above or 4°C lower than the temperature control setting, The Airway Temp LED will flash and audible alarm.	Please check the temperature probe and the water in the circuits.
The alarm tolerance of Chamber Temp. is above or lower than 4°C the temperature control setting, The Chamber Temp. LED will flash and audible alarm.	Please check the aluminum tray and temperature probe.
The Temperature probe disconnect or malfunction, the LED will flash and audible alarm.	Please confirm that the tubing is installed correctly.
The heated wire adaptor disconnect or malfunction, the LED will flash and audible alarm.	Please confirm that the tubing is installed correctly.

10. Disposition

Please DO NOT dispose VH-3000 as normal trash.
Please dispose it according to the local rule of the
disposition of electronic device/accessory waste.

11. Clean

The heated plate may be wiped with a damp cloth with
using clean water or 75% ethyl alcohol.

12. Maintenance

Check the power cable for damage and replace if it is
necessary.

Ensure heater plate surface is clean and free from pitting,
gouging etc.

13. Warnings

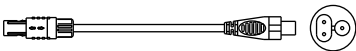
- 1.The product is intended for use by qualified medical staff with medical gloves.
- 2.Please follow waste disposal guideline locally.
- 3.The device is intended to be used in a professional healthcare facility environment e.g. hospitals.

Cautions

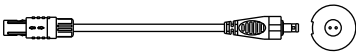
- 1.Please do not touch the heater plate while the device is working.
- 2.While moving, maintaining, serving or cleaning the device, please beware of the temperature on heater plate, no matter the device is or not in operation.
- 3.The performance and the safety of the device would be affected if the device is used with the consumables which are not manufactured by VADI Medical Technology Co. Ltd. Yangmei.
- 4.Use any blanket to cover the breathing circuit is not allowed.
- 5.NO MODIFICATION of this humidifier is allowed by user.
- 6.DO NOT try to dismantle or/and repair the humidifier by yourself if it is malfunction. Please contact local distributor for the service.
- 7.To avoid the risk of electric shock, this equipment must only be connected to supply mains with protective earth.
- 8.The AC power plug is served to easily connect and disconnect the humidifier. Do not use the receptacle that is difficult to disconnect the humidifier.
- 9.Serious incident in relation to this product must be reported to VADI, distributor and applicable competent authorities.

14. Optional Accessories

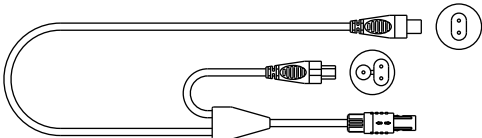
(A) VH30100
Power adaptor
(inspiratory heated)



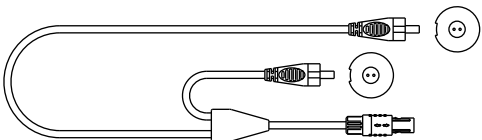
(B) VH30100L
Power adaptor
(inspiratory heated)



(C) VH30200
Power adaptor
(dual heated)



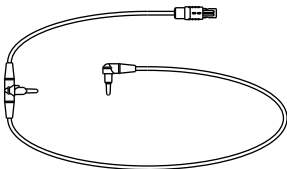
(D) VH30200L
Power adaptor
(dual heated)



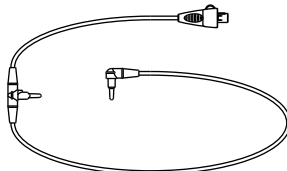
(E) VH30C
Connector
(2P/3P)



(F) VH30T150
Temperature Probe - 150cm



(G) VH30T165
Temperature Probe - 150cm



(H) 900110L

Heated wire (with port) - 110cm



(I) 900130L

Heated wire (with port) - 130cm



(J) 900150L

Heated wire (with port) - 150cm



(K) 900110LE

Heated wire (without port) - 110cm



(L) 900130LE

Heated wire (without port) - 130cm



(M) 900150LE

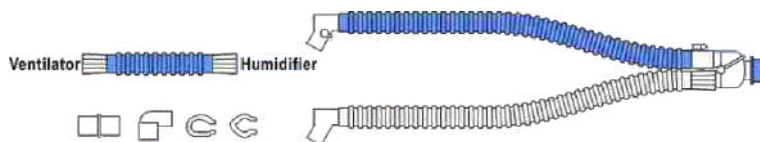
Heated wire (without port) - 150cm



Breathing Circuit:

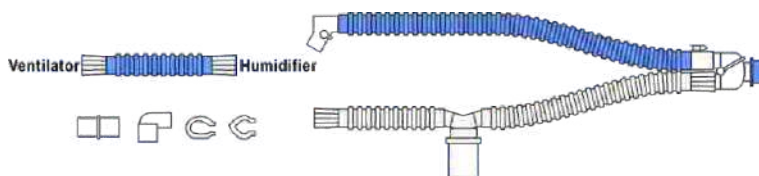
(A) G-214000

Disposable adult circuit (150cm/ 22mmID), dual heated, w/o chamber



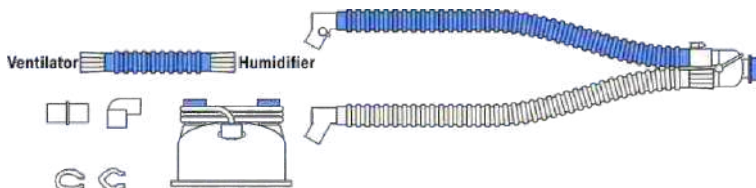
(B) G-215000

Disposable adult circuit (150cm/ 22mmID), inspiratory heated, w/o chamber



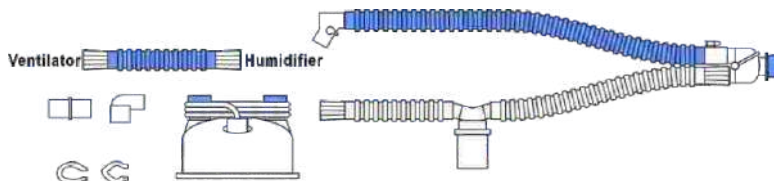
(C) G-216000

Disposable adult circuit (150cm/ 22mmID), dual heated, w/ chamber



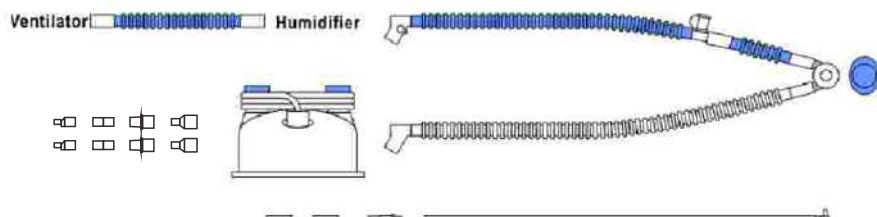
(D) G-217000

Disposable adult circuit (150cm/ 22mmID), inspiratory heated, w/chamber



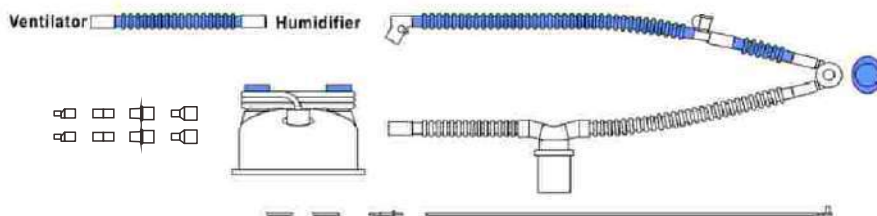
(E) G-218000

Disposable infant circuit (150cm/ 12mmID), dual heated, w/chamber



(F) G-219000

Disposable infant circuit (150cm/ 12mmID), inspiratory heated, w/chamber



Chamber

Item number	Description
G-314001	Disposable humidifier chamber
G-314001-0	Disposable humidifier chamber
G-314001-B	Disposable humidifier chamber
G-314001-C	Disposable humidifier chamber
G-314003-P	Disposable humidifier chamber
G-314002	Reusable humidifier chamber
G-314003	Reusable humidifier chamber
G-314004	Reusable humidifier chamber
G-314005	Reusable humidifier chamber
E-4011-07-2	Accessories-Absorbent paper for G-314004
E-4011-07-3	Accessories-Absorbent paper for G-314005

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